Data inputs

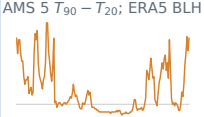
Hourly CH_4
WBEA, 12 stations



Wind
speed and direction



Stability
AMS 5 $T_{90} - T_{20}$; ERA5 BLH



Network geometry
station pairs (n = 21)



(a) Analytical framework

Quality control and eligibility → Temporal partitioning → Background adjustment → Prespecified estimators

$$R_i(t) = X_i(t) - \text{median}_{k \neq i,j} X_k(t)$$

H1
Stability association

$$y_t = \beta S_t + \gamma \log(U_t + 0.5) + \alpha_{c(t)} + \varepsilon_t$$

$c(t)$: year × month × hour

H2
Directional coupling



$$\rho_\ell = r[\Delta R_i(t-\ell), \Delta R_j(t)] - r[\Delta R_j(t-\ell), \Delta R_i(t)]$$

$$D_{ij} = \Sigma_{\ell=1}^{3} w_\ell [\rho_\ell(A) - \rho_\ell(R)]$$

H3
Regime composition

$$\Delta r = \Sigma_g \Delta\pi_g \bar{q}_g + \Sigma_g \bar{\pi}_g \Delta q_g$$

$$s = \Sigma_k c_k t_k / \Sigma_k t_k^2$$

composition share

(b) Confirmatory analysis

Protocol frozen
after development

→ **Confirmation**
Jan–Sep 2026, single run

→
- ✓ Block-bootstrap CI (H1, H3)
- ✓ $p = \dfrac{1 + \#\{D^{(k)} \geq D\}}{K + 1}$, $K = 246$ (H2)
- ✓ < 50% valid CH_4: station excluded

(c) Methodological evaluation (post-confirmation)

Synthetic calibration and falsification

$\hat{\alpha} = \#\{p < 0.05\} / n$ under null scenarios

power; advected-signal falsification

Season-restricted shift references

$U^{(k)}(t) = U(a_b + (t - a_b - 24k) \bmod L_b)$

shifts within calendar-based seasonal blocks b

Outputs

H1
$\hat{\beta}$ with 95% CI
ppm per SD of S

H2
D and finite-shift p
pooled over eligible pairs

H3
$\hat{s}$ with 95% CI
composition share

Decision
- Confirmed
- Not confirmed
- Not evaluable